AIORI-3 HACKATHON | CYBERSECURITY

TraceForge

Incident Timeline & Evidence Collection for Small Teams

PS-013 • Proposal Stage

Preserve log evidence, normalize timestamps to UTC, and reconstruct incidents with full source traceability.

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PROPOSAL STAGE

Abstract & Problem Description

Problem, proposed response, objectives, and project boundary

THE PROBLEM

Small IT teams often rebuild incidents by hand from separate host and application logs. Mixed formats, time zones, and clock offsets make event ordering unreliable, and manual review can miss critical context.

PROPOSED RESPONSE

TraceForge preserves original log evidence, records SHA-256 hashes, normalizes supported timestamps to UTC, and builds a searchable, source-linked timeline with an exportable incident summary.

PROJECT OBJECTIVES

- Ingest and parse two different log formats (host and application).
- Preserve originals; analyze normalized copies only.
- Expose source references, time corrections, and parsing gaps.
- Evaluate three synthetic scenarios against a manual triage baseline.

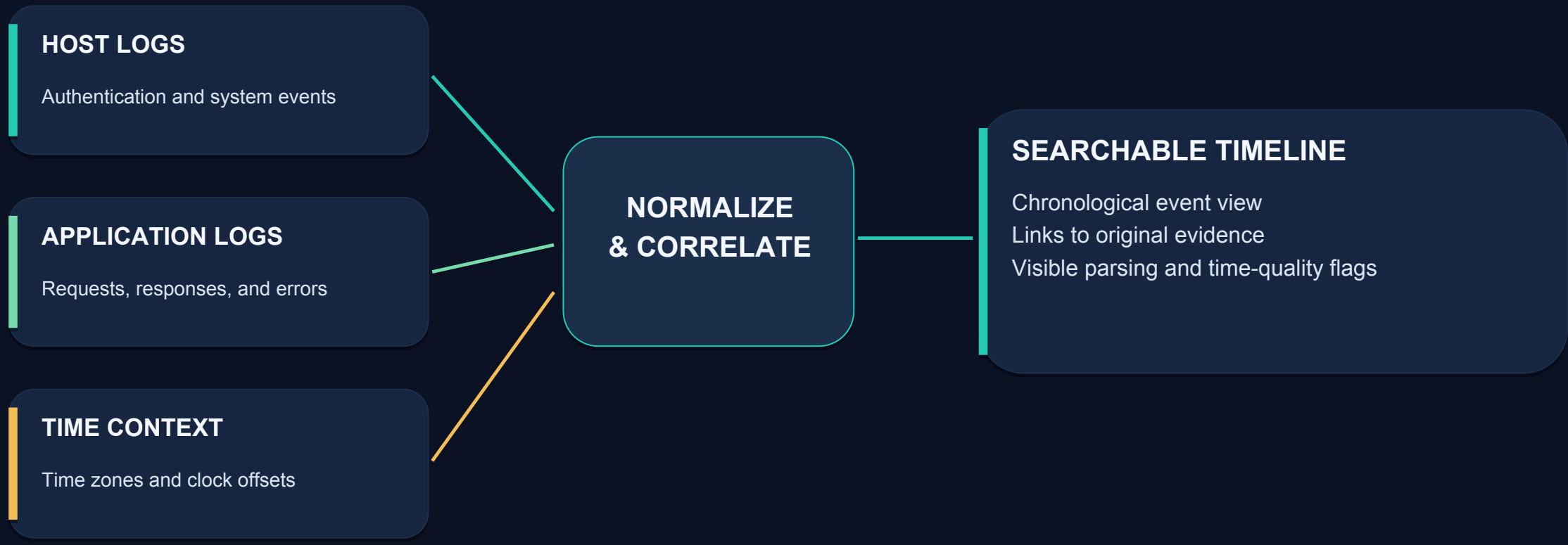
SCOPE BOUNDARY

In scope: offline log analysis and evidence reporting.

Out of scope: live endpoint response, malware removal, and forensic certification.

Background

Why a reliable incident timeline needs more than sorting log lines



DESIGN PRINCIPLE **Preserve the source, explain transformations, and expose uncertainty.**

Literature & RFC Review

Standards that shape TraceForge's evidence handling and time model

NIST SP 800-86

Guide to Integrating Forensic Techniques into Incident Response.

Informs evidence handling and the investigation workflow.
csrc.nist.gov/pubs/sp/800/86/final

IETF RFC 3339

Date and Time on the Internet: Timestamps.

Informs timestamp representation and parsing when time-zone information is present.
rfc-editor.org/rfc/rfc3339

OUR DESIGN RESPONSE

01 Traceability

Event → source record

02 Time transparency

Original → UTC + notes

03 Visible uncertainty

Gaps stay visible

TraceForge combines evidence preservation, timestamp normalization, and source traceability in one lightweight workflow for small teams.

Proposed Solution

Evidence-first processing with transparent timestamp handling

01 Collect
Two supported log formats

02 Preserve + hash
Original + SHA-256

03 Parse
Extract supported fields

04 Normalize
Valid timestamps to UTC

05 Correlate
Search + sort events

06 Report
Summary + gaps

PROPOSED EVENT RECORD

Source: original file / record reference
Time: original timestamp + normalized UTC
Integrity: evidence hash reference
Quality: parse status, time correction, uncertainty

PSEUDOCODE

```
for each collected file:
  preserve original; calculate SHA-256
  parse supported records
  if timestamp is valid:
    normalize to UTC
  else: flag uncertainty
  save event + source reference
build timeline; export summary
```

Every event keeps a reference to its source record and evidence hash.

Solution Optimization

Workflow comparison and a measurable evaluation plan

TASK	MANUAL WORKFLOW	TRACEFORGE
Log review	Open and inspect files separately	View supported sources in one timeline
Time handling	Compare time zones manually	Normalize valid times; retain original values
Source checking	Return to files to locate events	Follow event-to-source references
Parsing issues	May be overlooked	Record and surface parsing gaps
Summary	Compile notes manually	Export a structured incident summary

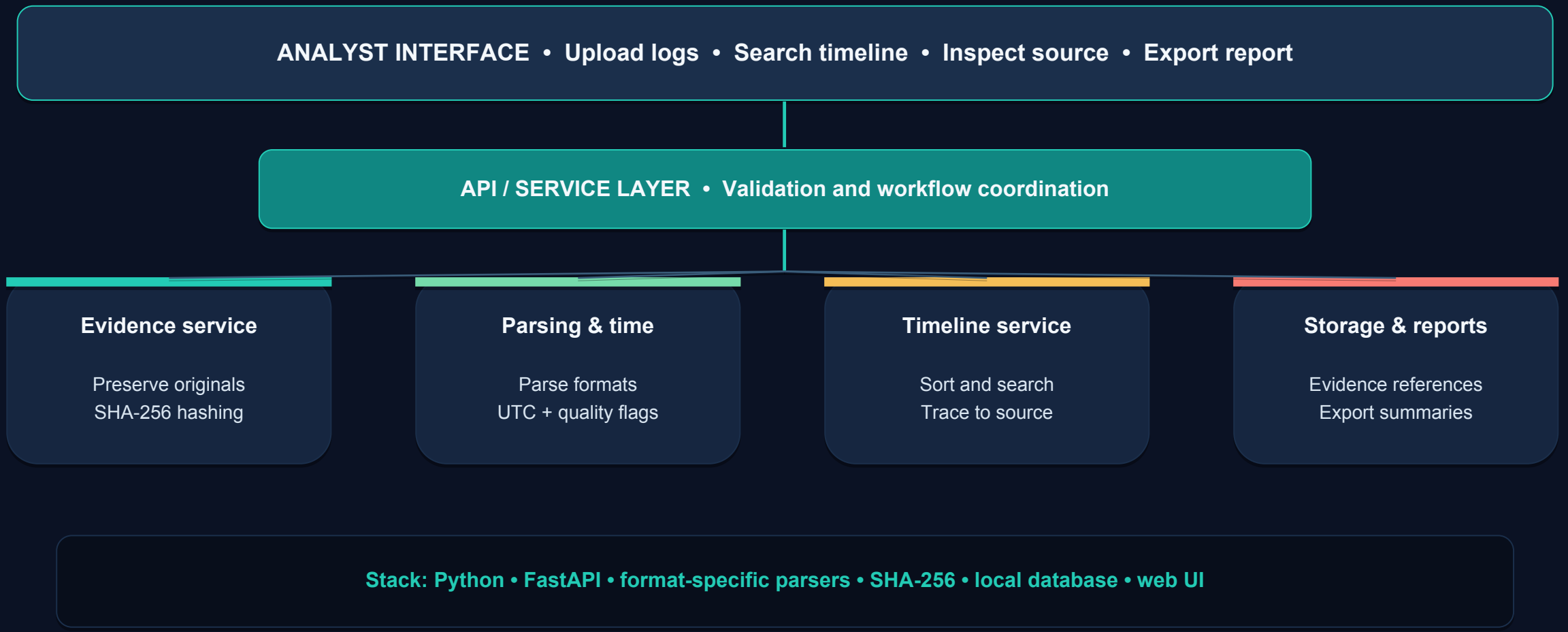
EVALUATION MEASURES

- **Triage time**
- **Parse coverage**
- **Hash verification**
- **Timeline correctness**

Manual and TraceForge runs use identical synthetic scenarios with the same defined start and end points.

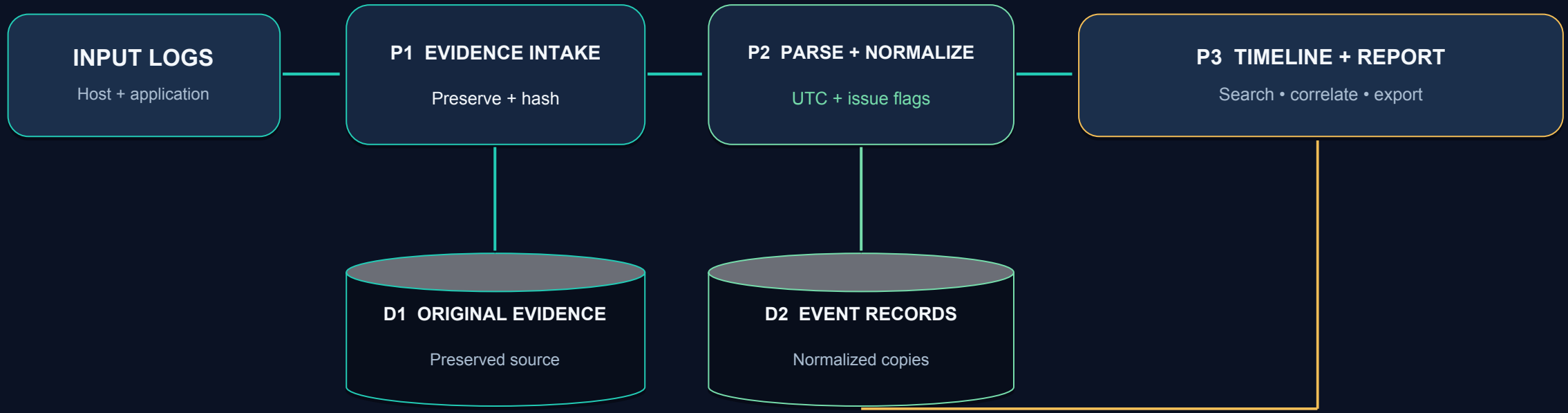
Solution Architecture & Design

Component view of TraceForge



Data Flow Diagram

Original evidence and derived records remain logically separate



CONTROL: retain source references • record parse failures • never silently discard ambiguous timestamps

Implementation Roadmap

Phased build and validation plan

PHASE 1

Scope & datasets

Formats, requirements, synthetic logs

PHASE 2

Evidence handling

Preserve originals; implement hashing

PHASE 3

Parsing & time

Parse fields; normalize UTC; record uncertainties

PHASE 4

Timeline & reports

Search, traceability, report export

PHASE 5

Test & validate

3 scenarios; manual comparison; document gaps

VALIDATION SCENARIOS

01 **Failed authentication**

02 **Correlated web activity**

03 **Mixed time zones + malformed records**

Bibliography

Primary references

[1] NIST SP 800-86

National Institute of Standards and Technology. Guide to Integrating Forensic Techniques into Incident Response.

https://csrc.nist.gov/pubs/sp/800/86/final

[2] RFC 3339

Klyne, G. and C. Newman. Date and Time on the Internet: Timestamps. IETF.

https://www.rfc-editor.org/rfc/rfc3339

[3] PS-013

AIORI Hackathon 3. Incident Timeline and Evidence Collection for Small Teams.

Problem statement, AIORI-3 Hackathon



THANK YOU

From scattered logs to a traceable incident timeline.

PRESERVE • NORMALIZE • CORRELATE • EXPLAIN

TraceForge | AIORI Hackathon 3 | PS-013